

Sonic Shower: An interactive sound installation exploring sonification of water

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Figure 1: Physical setup of Sonic Shower as an interactive installation.

Abstract

Sonic Shower is an interactive sound installation. The artwork relates to Velasco's research on distinguishing water temperature by its sound and the use of sound waves to cleanse the body as depicted in the science-fiction series *Star Trek*. In addition, Sonic Shower is a speculative design that imagines a dystopian future where individuals, confronted with severe water scarcity, employ

technology to replicate the shower experience for psychological solace. The installation creates a playful experience by converting a classic wall-mounted standing multi-shower system into loudspeakers and MIDI controllers that generate sound using a Max patcher.

CCS Concepts

• **Applied computing** → **Sound and music computing**; • **Human-centered computing** → *Sound-based input / output*.

Keywords

Data Sonification; Interactive Sound Installation; Shower Experience; Water Conservation

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1 Introduction

Showering, a primary human activity that involves water, considers the temperature of the water, spray intensity and direction, and sensory immersiveness, to be important factors. Velasco, in 2013, indicated that people can tell if water is hot or cold by listening to the sound it makes when it is poured into a container. [Velasco et al. 2013]. Ideally, people should perceive the temperature of the water through auditory characteristics during showers. However, temperature-related sound variations were rarely mentioned when people describe or recreate a shower experience.

Creative techniques allow artists to reimagine the experience of showering. As a starting point for this project, the authors speculate about a shower experience that completely replaces water with sound. Meanwhile, information such as the temperature of the water can be presented in an auditory experience using sonification. Max (Cycling '74), a visual programming language for music and multimedia commercialised since 1980s, is popular among researchers and artists to create recordings, performances, and installations. With this technology, computer-generated sound effects can be used in place of water to provide humans with a new audio-only shower experience.

This work addresses the following research questions.

RQ1: How to design a playful shower experience using the digital sound generated by the computer?

RQ2: How does a sound-only shower experience affect participants' thinking of showers?

To explore these questions, the authors designed the Sonic Shower, an interactive sound installation. The installation consists of two parts: a classic wall-mounted standing shower and an acrylic base. The installation retains the original appearance of the shower and the way of operation. However, all parts, including the showerheads, the faucet, and the drain, were replaced with loudspeakers. The user can freely move the showerhead around her/himself to create a spatialized sonic experience. This kind of immersivity of water-showering is an important aspect for this project. The three original faucet handles were transformed into MIDI controllers equipped with Arduino accessories, linked to the Max patcher on a laptop. This enables the participant to stand on the shower base and manipulate the sound emitted from the showerhead. Participants need not concern themselves with becoming wet, as the whole experience is a futuristic dry shower.

2 Background

In the 21st century, showering was more than an action for carrying out moral obligation and hygienic convention [Shove 2003]. Researchers concentrate on enhancing the comfort of the showering experience and addressing the environmental concerns related to water usage in showers. Generally, a five-to-ten-minute shower is the most satisfying [Adeyeye and She 2015]. Showers shorter than five minutes can be rushed and unsatisfying, while showers longer than ten minutes can be tiring and torturous. Excessively long showers, whether refreshing or relaxing, are not recommended in terms of water conservation. In addition to consciously controlling

the length of showers, the use of water-saving appliances is widely acknowledged as the best way to conserve water unconsciously [Batchelor et al. 2014]. In Hong Kong, research on the Voluntary Water Efficiency Labelling Scheme (WELS) for showers predicted water-saving appliances can lead to significant reduction in water consumption, energy use, and decline in carbon dioxide emissions [Wong et al. 2016]. However, it still takes improvements to economic incentives (e.g., water pricing), technical improvements (e.g., water-saving household appliances), or policy instruments and regulations to make the idea of minimising water consumption in showers more widely accepted [Koop et al. 2019].

2.1 Shower In Future

In the future, showering may change due to advances in technology or lack of water resources. In the science fiction series *Star Trek*, a method of cleansing the body with sound waves is demonstrated, applied to scenarios such as spaceships where water is extremely limited [Robert 2024]. In *Star Trek*, sound waves remove stains from the body by creating vibrations. In the actual world, the International Space Station (ISS) outlines the current methods that primarily involve the use of chemicals and extremely frugal water consumption [Wong 2017]. However, it motivated the authors to expand this project by speculating about how people might find spiritual comfort in a future with limited water supplies.

2.2 Related Work

In the context of the creative industry, artists produced works to promote the concept of conserving water during showers. Presenting at the International Conference on Auditory Display in 2013, INFODROPS [Tünnermann et al. 2013] is a sonification of data from the sound of water droplets falling into the bathtub and data from the water flow inside the pipe. It introduced a novel method to maintain awareness of water and energy consumption during showers using the unobtrusive and effective ambient auditory information display.

Meanwhile, "The Singing Shower," an interactive shower interface that responds to melody was presented at Sound and Music Computing Conference in 2022 [Seznec and Pauletto 2022]. With this system, users can sing melodies to control the flow of water, transforming the shower into an interactive musical instrument. By correlating water flow to user input and combining physical interaction with sound-responsive digital control, the design of this prototype promotes sustainable water use.

Besides, Sonification of water temperature are widely used as a method to address environmental issues. Norwegian jazz musician Håkon Kornstad sonified the ocean temperature of Barents Sea from 1993 to 2022, to raise awareness of the invisible changes happening in the ocean [Ocean 2022]. Damon Hart-Davis sonified the nearly decade-long monthly monitoring data of the water temperature from his home kitchen faucet and published it on Earth Notes, a website that promotes the concept of "cutting carbon at home to slow climate change" [Hart-Davis 2024]. "The Sea Surface Temperature Aria" by Dr Lee de Mora links sea surface temperature to musical instrument sounds consistently in the scale of A minor harmonic, which carries an uneasy and sorrowful tone that evokes a sense of dread and discord in the listener [de Mora et al. 2020]. These

sonification projects utilize temperature data collected for environmental monitoring purposes, allowing the audience to perceive environmental changes over annual timeframes both auditorily and emotionally. Although Sonic Shower also employs sonification technology for water temperature data and centers its theme on environmental issues, the audience cannot perceive environmental changes through it. Instead, in our work, sonification serves as a technical means to replace water with sound. The presentation of environmental issues at the conceptual level is achieved through speculative design.

2.3 Research of hearing temperature

Similar to vibrating solids, dripping and splashing initiate a deformation, which the restoring forces counteract. The formation and change of liquid surface resonant cavities determines the sounds [Gaver 1993]. When water is poured into a container composed of one of four materials (glass, porcelain, paper, or plastic), people can determine whether the water is hot or cold just by listening to the sound it makes [Velasco et al. 2013]. The difference could be determined from the sample spectrogram of acoustic stimuli.

3 Implementation

Sonic Shower is a sound interactive installation consisting of classic-style wall-mounted shower equipment in silver and a bathroom base in acrylic on the floor. Audiences can stand on the base and participate by twisting the valves, similar to the initial operation of this shower. The showerhead emits a computer-generated sound effect instead of water. Turning the two valves next to the faucet at different angles result in a sound experience that simulates different sensations of water temperature. Turning the centre valve switches the activated loudspeakers between the top shower, hand shower, and bottom faucet. Users can activate a maximum of one loudspeaker simultaneously. The sound is initially sharp and will become richer over time. After about five minutes of continuous use, the user should reach the optimal experience. The sound from the faucet or showerhead will instantly stop when the participant shuts off the water valve. The sound from the drain gradually diminishes, emulating the standing water running down the drain as the conclusion of a shower. The storyboard in Figure 2 will be printed on B5 paper and distributed to the audience at the exhibition as an instructional guide and invitation to participate.

3.1 Interaction and Installation

Technically, the interaction of Sonic Shower was realized using Arduino accessories as input and loudspeakers as output. The internal components of the original water valves were replaced with potentiometers, and wires passed through the water pipes to link the Arduino board located behind the wall. The initial input serial data is 0, increasing to a maximum value of 1024 after approximately two-thirds of a clockwise rotation. The handheld shower incorporates a three-axis accelerometer for capture its motion data.

These data would be imported into the Max software on the laptop to generate sound and select the appropriate loudspeakers to play. Four sets of loudspeakers were installed to replace the existing top shower, hand shower, faucet, and drain spout. Power amplifiers

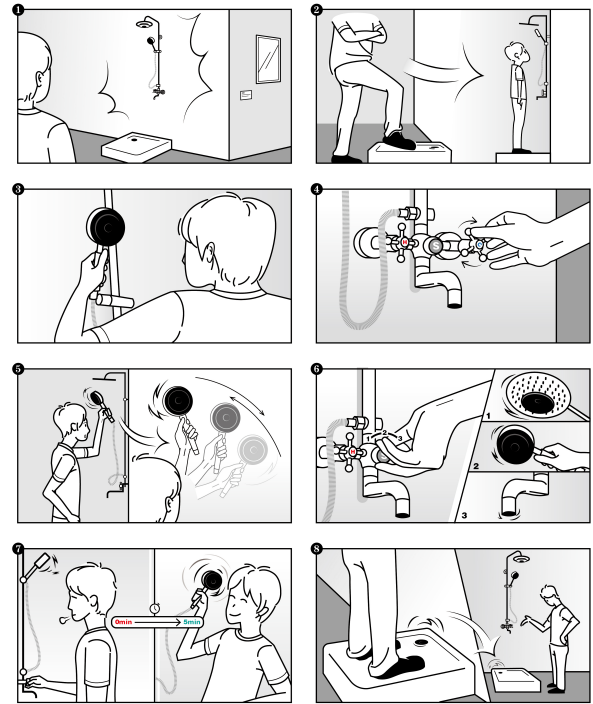


Figure 2: Storyboard of user guidance.

and audio interfaces were used to support the multi-loudspeaker system. More technical details are shown in Figure 3.

3.2 Sonification

In general, the sound is abstract and experimental, created using the software Max, the major sonification tool in this project. Initially, we recorded two clips of bathwater sounds in .wav format. The two pieces of audio are used as input, or the modulator signal in Max to be processed separately using ring modulation. Ring modulation is a signal processing function in electronics that mixes frequencies by adding two signals together to create a third signal [Emmerson 1977]. This method is believed to merge the information of temperature and movement into the sound in shower that initially exist. The variables are controlled by the serial input from potentiometers, which alters the pitch and speed of the input audio and, crucially, shapes the carrier wave. Besides, the component "munger~" is used to generate the simulated sound for the tap. "munger~" is a component based on real-time audio buffer cutting and reorganization and is believed to simulate the water sound characteristics emitted from the tap. We use "cycle~" as the source to control the output sound by adjusting the parameters of grain separation and grain pitch variation.

During pre-production, we referenced the method developed by [Wirtz 2006] for monitoring the temperature of water in a faucet with J-type thermocouple. However, due to the limited accuracy of the thermocouple used and the fact that the water temperature is influenced by environmental temperature, regional water supply,

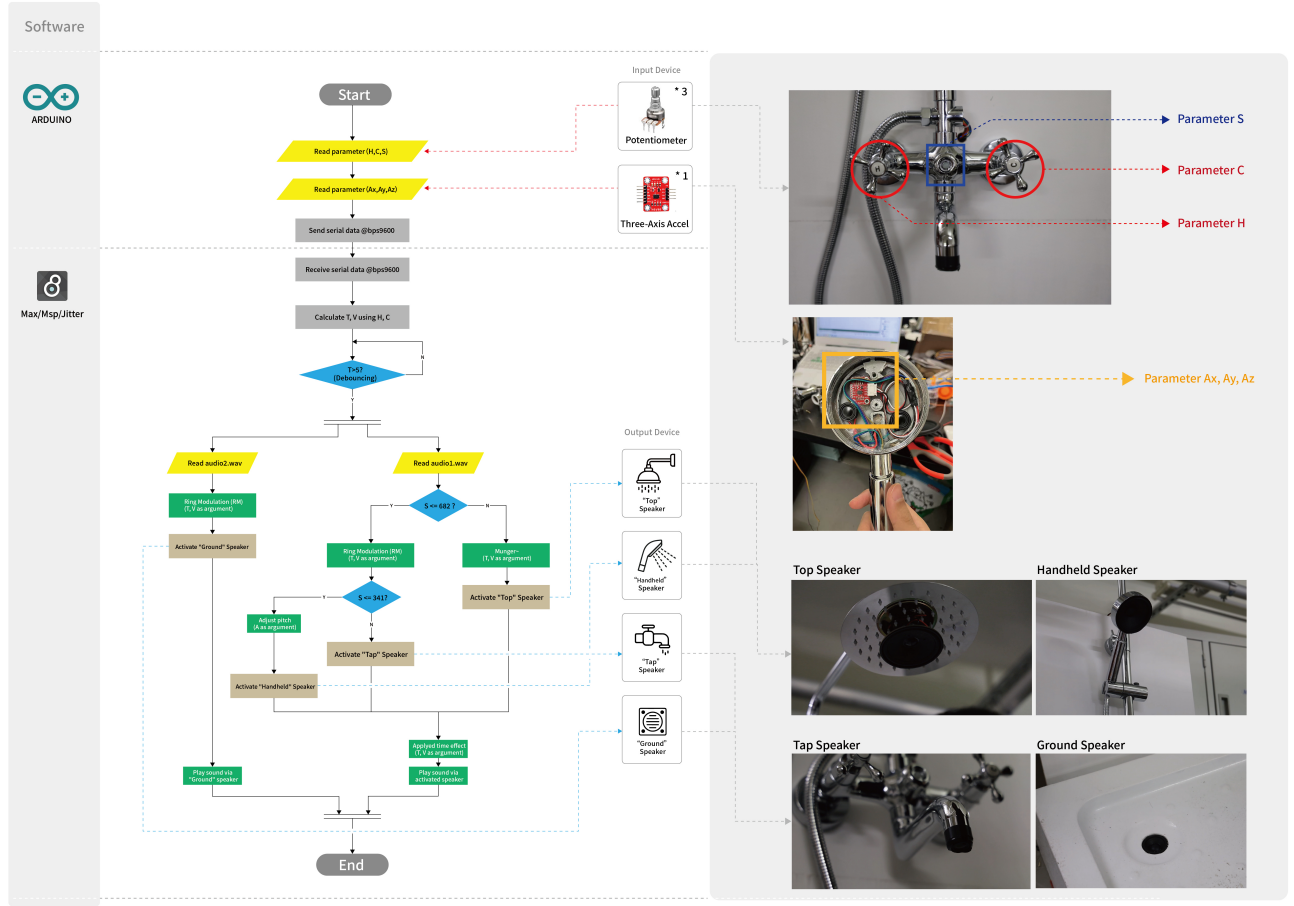


Figure 3: Flowchart of the interactive system, with photos of Arduino accessories and loudspeakers.

and pipe materials, the result in pre-production is not statistically significant. We can only estimate the output water temperature as a weighted average of the supplied cold and hot water temperatures. Two parameters, hot water (H) and cold water (C), were utilized, with a range of 0-1024, as read by the potentiometers in the shower knob, to calculate the simulated water temperature and flow rate. We assume that the supplied hot water temperature is 50 degrees Celsius and the supplied cold water temperature is 20 degrees Celsius, with a flow rate range of 0 to 100%. We apply the following formula to calculate the simulated water temperature (T) and flow rate (V), two variables that directly influence sound generation, to standardize the criteria for sonification.

$$T = \frac{H \times 50 + C \times 20}{H + C} \quad (1)$$

$$V(\%) = \frac{H + C}{2048} \times 100\% \quad (2)$$

We can identify the difference between the sonification of hot water and cold water in the spectrograms, as shown in Figure 4.

To match the real-life experience, we designed a showerhead motion effect, a fade-out effect, and a time effect. The showerhead

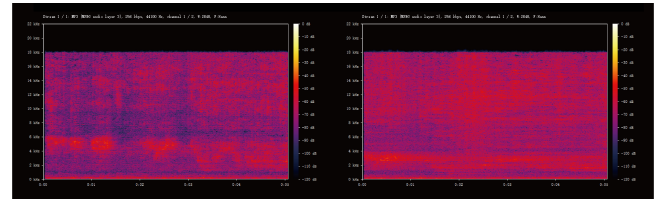


Figure 4: Spectrograms of the sonification. Simulated cold water with H = 0, C = 612 (left). Simulated hot water with H = 612, C = 0 (right).

motion effect alters the sounds of the handheld shower based on its movement. We installed a three-axis accelerometer sensor in the handheld showerhead (with parameters Ax, Ay, and Az) to capture the showerhead's motion data and map it to variables influencing the sound. The fade-out effect allows the sound to continue for an extended period after the water source is turned off. This effect only works on the drain loudspeakers that simulate the water slowly flowing into the drain after a shower. The time effect influences the sound over time on echoes. The effects of echo increase in intensity

at a rate determined by the variables T and V. It reaches its highest value after 5 to 10 minutes. The sound in general will be sharp for the first five minutes, when the echo isn't recognizable. It will be mild for 5–10 minutes, but listening after the peak will make the audience dizzy. Which refers to the study we discussed in the previous session that suggests the ideal shower length is 5 to 10 minutes.

In conclusion, the sonification is designed as an interactive shower experience system constructed with computer-generated sound. Unlike regular showering experiences, individuals stop adjusting the water temperature once it is suitable. This system enables users to perpetually manipulate the knobs via dynamic sound field reconstruction technology. Users are expected to perceive variations in sound corresponding to different water temperatures and flow rates, along with unique auditory sensations resulting from various locations where water emanates, including the vibrations transmitted to their feet when the drain loudspeakers are activated. The open interaction mechanism converts the "goal-attainment" function of regular bathing into a "process-exploration" auditory experience.

4 Discussion and Future Work

This artwork has been exhibited at the 19th SIG Playful Media Showcase. It is mounted on a wall in the gallery as shown in Figure 4, allowing every passing viewer to step forward and interact with it. The audience can voluntarily fill out a questionnaire containing 5 rating scale questions (1-5) about personal feelings during the experience and two open-ended questions about their expectations of future showers. Seventeen valid questionnaires were collected. The author also interviewed audience members ($n = 3$) who visited the exhibitions to understand their perceptions of this sonic shower experience and gain opinions on how sound could affect showers.



Figure 5: Sonic Shower presented as an art installation at the 19th SIG Playful Media Showcase

4.1 Qualitative Findings

The majority of the audience rated the experience as non-relaxing, with an average score of 2.3 on the Suffering-Relaxing Scale. The

duration of the experience predominantly ranged from 2 to 3 minutes, which prevented the time effects we set in sonification from being properly represented. The abstract style of sound may have challenged the exhibition audience's ability to maintain sustained concentration on the piece. However, the audience responded positively to the questions categorized as boring-fun, realistic-fantastic, and complex-easy to manipulate. Most audience engaged with the valves more as a playful activity by adjusting them back and forth. Viewers shown neutrality regarding their perception of warmth or cold. After an intentional reminder, they were able to immediately associate the difference they previously perceived with the warm/cold relationship.

Interviews revealed that the audience recognized the significance of water sound as a crucial element in the composition of the shower experience. One interviewee described this project as a solo auditory experience that blocks the senses of touch, warmth, and sight. "In my real life, sound in the shower functioned as background noise, only draws my attention when others was using the shower" (P1). Another interviewee gave a positive expression of the abstract style since the idea of integrating sensory information with computer-generated audio enhances the sci-fi aesthetic for him. During the conversation, all three interviewees expressed approval of the concept of conserving water during showers. One expressed discomfort stated, "I feel sorry for individuals in the future who may only convince themselves they have been showering this way" (P3).

In summary, feedback from participant showed the limitations of the methods used in this project in terms of practical presentation. Proving that the objective of creating a playable experience and using sound to represent variations in warmth and coolness was successfully accomplished. Addressing the issue of how participants are affected by a sound-only shower experience. After this shower experience, participants started to consider the idea of water conservation as well as how sound can be used to convey multisensory information.

4.2 Potential Development

The display format of Sonic Shower could be enhanced or expanded in response to the negative results of the previous user interviews. The exhibition can be transformed into a performance while the physical setup remains the same. The showerheads in this performance will be designed to produce sounds of significantly different styles when directed at different parts of the body using techniques like image recognition or the wearable sensors. Such a performance would complement the roles of skin involved as a crucial surface for producing the sound. The audience can experience the entire, well-designed Sonic Shower during the performance with less obstacles. Nevertheless, it is inevitable that the immersion and enjoyment of exploring by themselves will be diminished. Alternatively, we can extend the playability of sonic shower with other interface objects that could be found during a shower, such as liquid shampoo, a loofah, or a shower puff.

4.3 Speculative Design

Sonic Shower could be developed as a speculative design to provide psychological relief in situations where bathing with water is not

feasible due to extreme circumstances. Taking sunlight in hospitals as a reference, an adequate and appropriate exposure to sunlight can have a positive impact on the health and well-being of people in a hospital setting [Seyedahmadi 2019]. However, for reasons such as ensuring air quality, some patient rooms, especially in the Intensive Care Unit (ICU), lack transparent windows for exposing patients to sunlight. In that case, the Blue Sky luminaires were designed for the patient's bed area to give the illusion of a skylight while providing adjustable light on the bed [Maniccia et al. 2017]. Software was written to set behaviors for various lighting scenes in the ICU throughout the 24-hour day to mentally support the patient's 24-hour rhythms. Similarly, Sonic Shower can be conceptualized as a mental reaction of shower. This could be implemented in airports to offer a dry solution for travelers who prefer not to change clothes or carry wet towels, while still seeking a psychologically relaxing experience akin to a shower. This approach may also offer psychological art therapy for individuals who cannot access water due to dermatological conditions like extensive burns. In these scenarios, the shower experience requires redesign to enhance practicality and comfort. In addition, Sonic Shower represents a depiction of a dystopian future. In a technologically advanced yet impoverished future characterized by water scarcity or monopolization, individuals find solace in computer-generated auditory simulations of showers. This approach addresses the audience's awareness of water wastage and the issue of totalitarianism.

Different from the other sonification projects mentioned earlier, the narrative of the Sonic Shower is not achieved by directly allowing the audience to hear data representing natural changes. Instead, in this project, sonification is used as a method to recreate the immersiveness of a water-based shower within an audio-only shower experience. Its concept of advocating water conservation is realized through speculative design. As an art project centered on environmental protection, Sonic Shower contributes by offering a possible approach that applies sonification to daily scenarios and interprets its theme through speculative design. It also demonstrates an approach to sonification that does not rely on large datasets, prompting the audience to recognize the importance of water conservation in the process of reflecting on the future.

5 Conclusion

In conclusion, this paper demonstrates a new shower experience based on the sonification of water temperature. The approach demonstrated in the paper focuses more on expanding from real-life experiences than on restoring real sound effects, conveying to the audience an auditory difference that approximates the difference between cold and warmth. The aim is to create a playable shower experience that promotes exploration without compromising the immersion of the showering experience. The auditory experience in Sonic Shower represents a synthesis of multi-sensory data. It has been computerized to exceed the amount of information that the sound of water can originally carry. The wall-mounted shower functions as the main component of the installation, facilitating the immersive experience and reinforcing the project's narrative. Meanwhile, Sonic Shower represents a speculative design that envisions how humans could replicate the shower experience in a future characterized by water scarcity. This sound-only shower presents a

darkly humorous perspective on a dystopian future while satirizing current water wastefulness. Participants are expected to consider the significance of water conservation regarding future shower experiences following the project.

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